

High FREQUENCY TRADING WITH MULTIPLE AGENTS

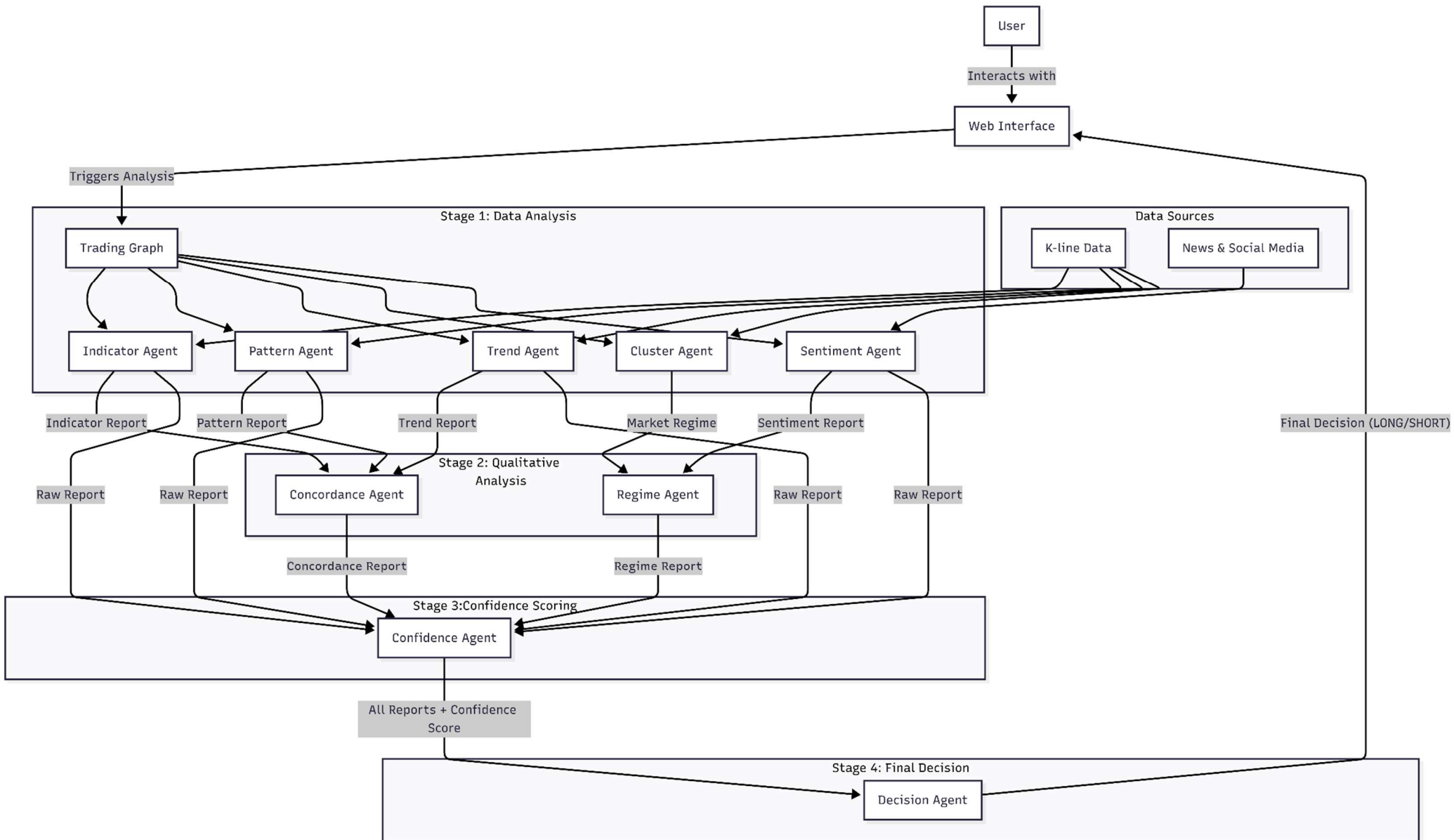


fig: High-level Architectural Diagram

DIAGRAM EXPLANATION

Stage 1: Parallel Expert Analysis (First Analysis)

- The user starts the process from the **Web Interface**.
- The **Trading Graph** activates all expert agents **at the same time**.
- These agents work independently, so the process is fast.

What each agent does:

- **Indicator Agent** → studies technical indicators from K-line data.
- **Pattern Agent** → finds chart patterns from K-line data.
- **Trend Agent** → checks whether the market is going up or down.
- **Cluster Agent** → identifies the current market condition (like trending or sideways).
- **Sentiment Agent** → analyzes news and social media to understand market mood.

All agents do their work in parallel, without waiting for each other.

Stage 2: Thematic Analysis (Combining Ideas)

- This stage starts **after Stage 1 is complete**.

Two main agents work here:

- **Regime Agent**
 - Combines:
 - Market condition (from Cluster Agent)
 - Market mood (from Sentiment Agent)

- Decides the overall market environment (good or risky to trade).
- **Concordance Agent**
 - Checks reports from:
 - Indicator Agent
 - Pattern Agent
 - Trend Agent
 - Verifies whether all technical signals are **agreeing with each other**.

This stage checks “Are things matching or conflicting?”

Stage 3: Synthesis & Confidence Scoring (Final Check)

- The **Confidence Agent** collects **all reports** from previous stages.
- It checks:
 - How strong the signals are
 - How well the agents agree with each other
- Then it creates a **Confidence Score**:
 - **High score** → signals are clear and aligned
 - **Low score** → signals are mixed or unclear

This score shows how reliable the trade idea is.

Stage 4: Final Decision (Action Time)

- The **Decision Agent** gets:
 - All agent reports

- The Confidence Score
- Based on this:
 - It decides **LONG (buy)** or **SHORT (sell)**
 - If confidence is low, it may act carefully or avoid risk
- The final result is sent back to the **Web Interface** for the user to see.